

AFS–ICIS-Air Crosswalk Brief

Script: `scripts/explore/10_frs-crosswalk.R`

Primary output: `afs_to_icis_crosswalk.csv`

Purpose: Link facilities in the legacy Air Facility System (AFS) to modern ICIS-Air records through EPA’s Facility Registry Service (FRS).

1 Overview

AFS and ICIS-Air describe facilities regulated under the Clean Air Act, but they do not use the same identifier. AFS uses a 10-character **AFS_ID**; ICIS-Air uses a program-system identifier called **PGM_SYS_ID**. The crosswalk uses the FRS **REGISTRY_ID** as a bridge:



Figure 1: The crosswalk joins AFS to ICIS-Air in two steps.

The primary crosswalk is **deterministic**: it uses exact identifiers supplied by EPA systems, not facility-name similarity or an inferred geographic match. A secondary investigation searches the facilities that fail the identifier crosswalk using exact, conservatively standardized facility fields. Those secondary results are retained as **candidates only**; they are not added to the crosswalk.

2 Identifiers and Units

Identifier	Meaning	Important nuance
AFS_ID	Legacy AFS air-facility identifier; 10 characters, including meaningful leading zeros.	An AFS record is a program record and is not necessarily a unique physical site.
PGM_SYS_ID	Program-specific identifier used in ICIS-Air and FRS program links.	One physical site can have multiple air-program IDs.
REGISTRY_ID	FRS identifier intended to represent a physical facility across EPA programs.	Several AFS or ICIS-Air IDs can share one REGISTRY_ID .

Table 1: Identifiers used by the crosswalk.

This distinction matters. The primary output answers: “Which ICIS-Air program records belong to the same FRS facility as this AFS record?” It does not always answer: “Which single ICIS-Air ID is the one-to-one successor of this AFS ID?”

3 Primary Identifier Crosswalk

Hop 1: AFS ID to FRS Registry ID

The FRS program-links file contains rows for many EPA program systems. The script first keeps records whose program acronym is AIR. Most AIR program identifiers are 18 characters long and contain the legacy AFS identifier in their final 10 characters:

0K0000004000300809 → 4000300809

The script therefore:

1. reads identifiers as character strings so leading zeros are preserved;
2. restricts FRS AIR records to the observed 18-character format;
3. extracts the final 10 characters; and
4. retains an extracted value only when it exactly equals an AFS_ID in AFS_FACILITIES.csv.

Why the final 10 characters are used

The script compares two extraction methods. The final-10-character rule matches 59.3% of all FRS AIR rows to a real AFS identifier. An alternative that removes a state prefix, strips leading zeros, and re-pads the result matches only 52.5%. The latter transformation can change meaningful digits and is therefore rejected.

The extracted value is the AFS identifier, not a coincidental numeric collision. This is confirmed by checking, for every matched pair, whether the AFS record and the ICIS-Air record the rule links it to describe the same physical facility. Across all 165,078 matched AFS_ID-ICIS-Air pairs:

- the two records agree on **state in 100.0%** of cases (the FIPS state code embedded in the extracted identifier never disagrees with the linked facility's state);
- the facility **names are exactly equal** (after upper-casing and removing punctuation) in 70.7% of cases, and **share a word of four or more characters** in 92.9% of cases.

A coincidental match between a random 10-character substring and the list of 236,734 real AFS identifiers could not produce perfect state agreement and name agreement at this rate. The roughly 7% of pairs whose names do not overlap are consistent with renaming or ownership change between the 2014 AFS freeze and the current ICIS-Air data (the state still agrees and the identifier match is exact), not with an incorrect extraction.

Why only 18-character AIR IDs are used

FRS AIR identifiers range from 6 to 18 characters. In the current file, 234,717 of 279,245 AIR rows (84.1%) have the 18-character structure. The shorter identifiers use other formats and do not reliably encode an AFS ID in the same position. Treating every AIR identifier as if it followed the 18-character layout would create false structure.

Hop 2: Registry ID to ICIS-Air

The ICIS-Air facilities table contains both `REGISTRY_ID` and `PGM_SYS_ID`. After Hop 1, the script joins the FRS registry identifiers to ICIS-Air directly. If one registry has several ICIS-Air IDs, the output retains all of them in a semicolon-separated field and records the number in `n_icis_ids`.

4 Primary Crosswalk Results

Measure	Result
Unique AFS facilities	236,734
AFS IDs linked to an FRS Registry ID	164,935 (69.7%)
AFS IDs reaching at least one ICIS-Air ID	164,880 (69.6%)
AFS IDs linked to multiple Registry IDs	208 (0.13% of linked)
Unmatched AFS IDs	71,799 (30.3%)

Table 2: Coverage of the deterministic identifier crosswalk.

Hop 2 is nearly complete once Hop 1 succeeds: 157,331 of 157,380 reached registry identifiers also appear in ICIS-Air. The main coverage constraint is therefore the availability of a recognizable legacy AFS identifier in the current FRS AIR program links.

5 Different Relationship Cases

Case 1: One AFS ID, one Registry ID, one ICIS-Air ID

This is the simplest case. The AFS identifier is extracted from an FRS AIR ID, the associated `REGISTRY_ID` appears once, and that registry has one ICIS-Air record.

Case 2: Several AFS and ICIS-Air IDs at one physical facility

FRS can assign several air-program records to one physical facility. For example, both `AFS_ID 0100300013` and `AFS_ID 0100300014` describe Gulf Coast Mineral at the same Foley, Alabama address. FRS assigns both corresponding ICIS-Air IDs to `REGISTRY_ID 110010380373`:

0100300013	→	110010380373	→	AL00000000100300013
0100300014				AL00000000100300014

Consequently, each AFS ID receives both ICIS-Air IDs in the facility-level crosswalk. This does not mean AFS ID 13 “became” ICIS-Air ID 14. It means FRS groups all four program records under one physical-facility record. A future record-level output could distinguish the suffix-preserving *direct ID* from *other ICIS-Air IDs at the same registry*.

Case 3: One AFS ID linked to multiple Registry IDs

Only 208 linked AFS IDs map to more than one registry. These cases are structurally ambiguous: they may reflect duplicate registry records, facility splits, moves, or another administrative history. The crosswalk preserves each `AFS_ID-REGISTRY_ID` pair rather than choosing one registry.

Case 4: Registry match but no ICIS-Air ID

Fifty-five unique AFS IDs reach FRS but do not reach an ICIS-Air program record. These records remain in the crosswalk with zero ICIS-Air IDs. They should not be treated as successful end-to-end links.

Case 5: No recognizable AFS identifier in FRS AIR links

For 71,799 AFS IDs, Hop 1 fails. A reproducible sample of 2,000 unmatched IDs was searched as a literal substring anywhere in the FRS AIR identifiers; none appeared. This establishes that their recognizable 10-character AFS identifiers are absent from the tested AIR ID field. It does **not** establish that the physical facilities are absent from FRS.

6 Why the Primary Identifier Match Is Incomplete

The data show several relevant patterns, but they do not identify one definitive cause for every unmatched record.

- **Operating status matters only modestly.** Operating facilities match at 73.3%; permanently closed facilities (X) match at 58.0%; planned facilities (P) match at 59.9%. This is consistent with incomplete migration or maintenance of older records, but it does not prove that mechanism, and it is far weaker than the state pattern below.
- **Coverage is strongly state-specific.** Identifier match rates are close to binary across states (Table 3) and are the dominant pattern in the unmatched set; see the subsection below.
- **Program records can be consolidated or split.** Changes in ownership, permitting structure, or facility organization can produce several historical and modern IDs for one site.
- **Legacy data can include obsolete or duplicate records.** Some AFS records may never have received a corresponding FRS AIR identifier.

These are explanations to investigate, not assignments made by the script. The primary crosswalk treats an AFS ID as unmatched whenever the exact identifier bridge is unavailable.

State, not facility status, governs coverage

Identifier match rates are close to binary across states. Among states with at least 1,000 AFS facilities, 12 of 38 match fewer than half of their facilities, while several match essentially all of them.

Lowest match			Highest match		
State	AFS fac.	% matched	State	AFS fac.	% matched
WV	1,021	0.0	CO	26,843	100.0
IL	20,197	0.6	OK	8,477	100.0
SC	4,428	1.4	VA	8,709	100.0
OH	6,602	1.6	NE	5,664	100.0
MI	3,609	5.0	KS	6,912	100.0
NC	7,807	7.9	GA	6,559	99.9
NY	12,379	11.3			

Table 3: Identifier crosswalk match rate by state (states with $\geq 1,000$ AFS facilities).

This is not explained by facility closure. Restricting to *operating* facilities, the same states still fail: 98.5% of operating AFS facilities in Illinois and 98.3% in Ohio are unmatched, versus near-zero in Oklahoma, Virginia, and Nebraska. Closure is only a weak factor overall (42.0% of permanently closed facilities are unmatched versus 26.7% of operating ones), and a majority of the unmatched — 45,213 of 71,799 (63%) — are operating. Whether an AFS identifier can be recovered from the FRS AIR field therefore depends primarily on the *state*, consistent with state-specific identifier formats or migration practices rather than facility-level characteristics.

Robustness: relaxing the 18-character restriction

A natural question is whether the low-match states fail because their AFS identifiers sit in *non-18-character* FRS AIR ids that the script discards. They do not. Extracting the final 10 characters of *every* AIR id of length ≥ 10 (not just the 18-character ones) raises the number of linked AFS facilities from 164,935 to 164,940 — **five additional facilities** — and recovers **zero** in the low-match states (Illinois, Ohio, West Virginia, South Carolina, New York all unchanged). The 28 non-18-character matches that do appear are all state-consistent (the embedded FIPS state matches the facility’s state in 100% of cases), so relaxing the rule is safe but immaterial.

The reason is visible once the FRS AIR ids are examined by state. For each state below: how many AIR rows FRS holds, the share that are 18 characters, and the share whose final 10 characters are an actual AFS identifier.

State	FRS AIR rows	% 18-char	% final-10 is a real AFS_ID
WV	1,393	0.0	0.0
IL	28,445	0.5	0.4
SC	3,436	1.9	1.8
OH	10,919	97.3	1.0
MI	3,489	97.2	3.8
NY	12,623	99.7	11.5
MT	218	100.0	88.5
LA	13,834	100.0	92.6
MD	11,119	99.8	98.9

Table 4: FRS AIR id format by state. Length is not the binding constraint; whether the embedded 10 characters are the AFS identifier is.

Three distinct failure modes appear, none fixable by changing the length rule:

1. **A different identifier format (e.g. Illinois, South Carolina, West Virginia).** The state is present in FRS’s AIR program, but its ids are not 18 characters and do not embed the AFS code. An Illinois AIR id is IL000031159AFL (14 characters; final 10 “0031159AFL” contains letters and is not an AFS plant code), whereas a real Illinois AFS_ID looks like 1700500056 (FIPS 17). Relaxing to non-18-character ids does not help because those ids still do not carry the AFS code.
2. **An 18-character shell with a different payload (e.g. Ohio, Michigan, New York).** The ids *are* 18 characters — they already pass the filter — but the embedded 10 characters are a different identifier. An Ohio AIR id is OH0000000744000180; its final 10 (“0744000180”) do not begin with Ohio’s FIPS code (39) and match no AFS_ID, while a real Ohio AFS_ID looks like 3911300378. For contrast, a working Louisiana id is LA0000002277700603 → 2277700603 (FIPS 22), which is a real AFS_ID.
3. **Sparse air-program coverage (e.g. Montana).** Where Montana ids do embed an AFS code it works (88.5%), but FRS holds only 218 AIR rows for Montana’s 2,602 AFS facilities — most are simply absent from FRS’s AIR program.

The binding constraint is therefore the *content* of the FRS AIR identifier, not its length. For the low-match states, FRS stored a non-AFS air-program identifier (or no air record at all), so no identifier-extraction rule can reach them; the name/address candidate search below is the only available path.

7 Secondary Candidate Investigation

Purpose

The secondary investigation asks a narrower question: “Does an unmatched AFS record have an FRS or ICIS-Air record with exactly agreeing facility information after harmless formatting differences are removed?” It does not alter the deterministic crosswalk.

Standardization decisions

The script applies only conservative formatting standardization:

- convert text to uppercase;
- replace punctuation with spaces;
- collapse repeated whitespace; and
- compare the first five numeric ZIP-code digits.

The script does **not** remove corporate suffixes, expand street abbreviations, correct spelling, infer renamed companies, geocode addresses, or calculate fuzzy similarity. Those operations require judgment and could create false matches.

Exact candidate rules

An unmatched AFS record becomes a candidate when it agrees with an FRS or ICIS-Air record on at least one of the following keys:

1. normalized facility name + state + ZIP5;

2. normalized street address + state + ZIP5; or
3. normalized facility name + city + state.

Each candidate records which rules produced it. Agreement on all three rules is stronger descriptive evidence than agreement on one rule, but the script does not convert that difference into an accepted-match threshold.

8 Secondary Investigation Results

Finding among 71,799 unmatched AFS IDs	Count	Percent
At least one exact FRS candidate	51,091	71.2%
Exactly one FRS candidate	40,957	57.0%
Multiple FRS candidates	10,134	14.1%
FRS candidate with an AIR program link	44,305	61.7%
FRS candidate with an ICIS-Air ID	44,304	61.7%
At least one direct ICIS-Air candidate	48,784	67.9%
No exact FRS or ICIS-Air candidate	15,462	21.5%

Table 5: Exact-after-formatting candidate-search results.

Among the unmatched records, 19,041 have exactly one FRS candidate and agreement under all three candidate rules. This is a useful high-evidence review group, but it remains unverified. The candidate file also contains many one-rule matches based only on a shared address or a shared name and location; these can be genuine, but can also represent headquarters, colocated businesses, reused industrial sites, or generic facility names.

9 How to Interpret Candidate Cases

- **One candidate; all three rules agree.** The records agree on current name, address, city, state, and ZIP. This is a high-evidence case to prioritize for manual review, but it should not be accepted automatically without an approved rule.
- **One candidate; address rule only.** The standardized street address and ZIP agree, but the facility names differ. This could reflect a rename or ownership change, or it could be a different entity colocated at the same address.
- **One candidate; name/location rule only.** The name and general location agree, but the address differs or is missing. This could reflect an address update, a move, or incomplete fields and requires review.
- **Multiple candidates.** The AFS record agrees with several registry or ICIS-Air records. The result is ambiguous; the script does not select one without additional information.
- **Candidate has an AIR or ICIS-Air link, but the AFS ID is absent.** The physical facility may exist in modern systems under a different air-program identifier. This is a potential migration or identifier-change case, not a proven successor link.

- **Candidate exists only in another EPA program.** FRS contains a similar facility but no current AIR or ICIS-Air record. This is useful evidence that the site exists in FRS, but it is not an AFS-ICIS-Air link.
- **No exact candidate.** No record agrees under the conservative keys. These cases may require fuzzy matching, historical sources, or geocoding, or they may genuinely lack a modern record.

10 What the Script Does Not Infer

- It does not declare that an exact name/address candidate is the same legal or regulated entity.
- It does not choose among multiple candidates.
- It does not assume that a shared `REGISTRY_ID` implies one-to-one program-record succession.
- It does not infer why an AFS identifier is missing from FRS.
- It does not add fuzzy, geospatial, industry-code, or temporal matches to the primary crosswalk.
- It does not replace the primary identifier crosswalk with secondary candidates.

11 Recommended Analytical Use

1. Use the deterministic `AFS_ID-REGISTRY_ID-ICIS-Air` crosswalk as the main linkage.
2. Preserve all ICIS-Air IDs associated with a registry when analysis is at the physical-facility level.
3. For record-level analysis, separate the suffix-preserving direct ICIS-Air ID from other IDs at the same registry.
4. Treat multiple-registry AFS IDs as ambiguous unless their histories are reviewed.
5. Use secondary candidate outputs for audit and manual review, not as accepted links.
6. Before adding secondary matches, pre-specify and validate an acceptance rule on a reviewed sample. Report deterministic and inferred links separately.

12 Output Files

- `afs_to_icis_crosswalk.csv`: deterministic crosswalk, one row per `AFS_ID-REGISTRY_ID` pair; `crosswalk_coverage.csv` contains coverage totals.
- `unmatched_afs_records.csv`: original unmatched records; the by-status and by-state files provide coverage diagnostics.
- The exact-FRS and exact-ICIS candidate files contain candidate pairs and evidence rules.
- The candidate-summary and investigation-summary files provide one-row-per-AFS and aggregate results.

13 Sources

- EPA ECHO, *AFS Data Downloads*
- EPA ECHO, *ICIS-Air Data Downloads*
- EPA ECHO, *FRS Data Downloads*
- Direct verification against the local AFS, ICIS-Air, FRS program-link, and FRS facility files.
- Results generated by `scripts/explore/10_frs-crosswalk.R`.